Introduction to Sampling and Reconstruction

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Key Concepts

1) Sampling is used to convert signals in the physical world with continuous independent variables (e.g., time) to a sequence of numbers that can be processed by a computer.

2) Reconstruction is used to convert a sequence of numbers into a signal with a continuous independent variable (e.g., time) for interaction with the physical world.

3) Sampling and reconstruction involve a mixture of continuous-time and discrete-time signals.

4) The Fourier transform is normally used to analyze mixtures of continuous-time and discrete-time signals.

5) Important questions to be addressed in subsequent lessons:

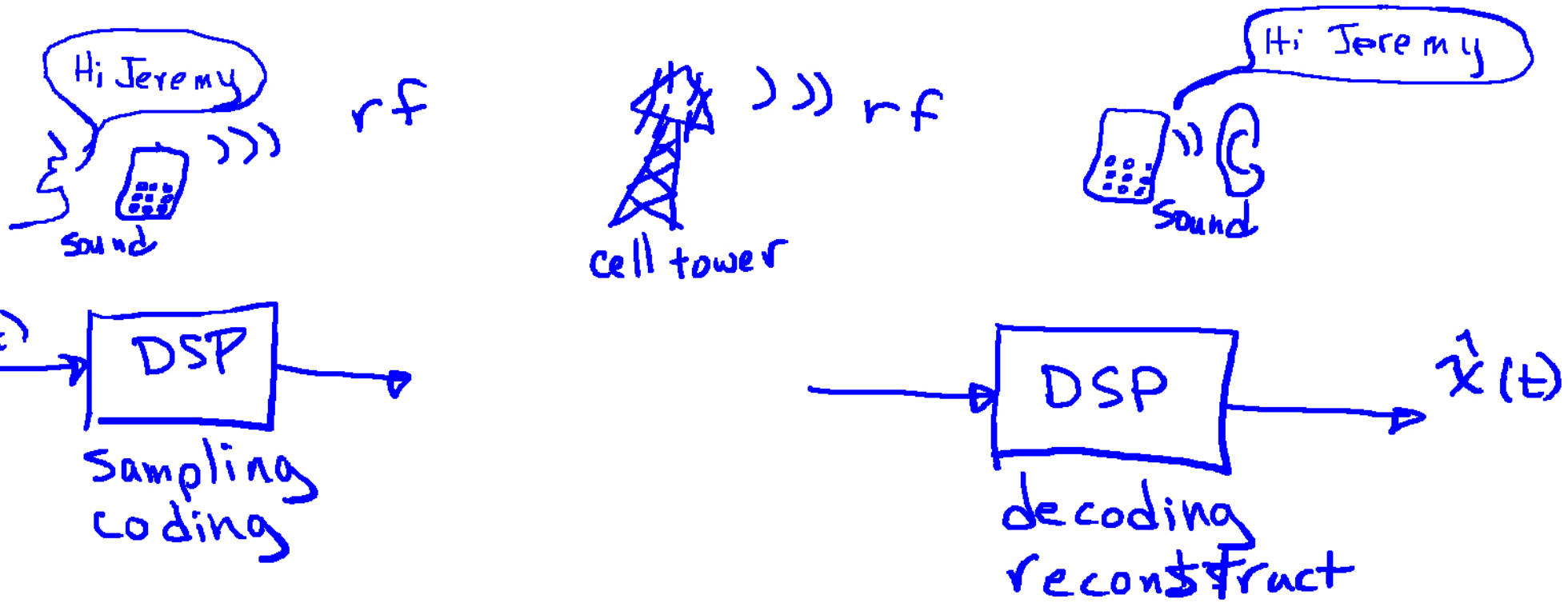
    a) How frequently should one sample a signal?

    b) What operations should be used to convert a discrete-time signal back to continuous time?

    c) How does sampling affect the signal-processing system?

Video Screenshots

Intro to Sampling/Reconstruction



"Mix" of continuous- and discrete-time signal

Analyze Sampling using FT

1) Relate ω to Ω
2) Relate FT of sampled signal to FT of original

How fast to sample?

How to reconstruct?

Specifications on discrete-time filter?

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